

Summary

This topic synthesizes the principles of Minimal Cost Spanning Trees (MCST) using greedy algorithms. The goal of an MCST is to select a subset of edges in a weighted graph that connects all vertices without cycles while minimizing total weight.

- Brute Force Limitation: Complete graphs have $(n^{\{n-2\}})$ spanning trees, making $O(n^n)$ searches infeasible.
- Greedy Solutions: Kruskal's Algorithm (edge-based, may form a forest temporarily) and Prim's Algorithm (vertex-based, always connected).
- Negative Weights: Both algorithms function correctly with negative edge weights.
- Uniqueness: Distinct edge weights yield a unique MCST; equal-weight edges may result in multiple MCSTs with identical total cost.

I. Fundamental Definitions and Spanning Tree Properties

A spanning tree $(G' = (V, E'))$ of graph $(G = (V, E))$ must satisfy:

- Vertex Inclusion: All vertices are included.
- Edge Count: Exactly $n-1$ edges for n vertices.
- Connectivity: All vertices are connected.
- Acyclicity: No cycles.
- Weighted Graphs: MCST minimizes the sum of edge weights.

II. Quantitative Analysis and Counting Spanning Trees

1. Complete Graphs

For (K_n) , number of spanning trees = $(n^{\{n-2\}})$.

Brute-force complexity: $O(n^n)$, which is not computationally feasible.

2. Kirchhoff's Theorem (Matrix Tree Theorem)

For non-complete graphs, the number of spanning trees can be computed using an adjacency matrix:

- Replace non-diagonal 1s with -1.
- Replace diagonal 0s with vertex degree.
- Number of spanning trees = determinant of any cofactor/minor.

Example: A sample graph with vertices A, B, C, D produced 8 spanning trees using Kirchhoff's method.

III. Kruskal's Algorithm: Edge-Based Greedy Approach

Methodology

1. Sort all edges in ascending order of weight.
2. Pick the edge with minimum weight.
3. Add it to the spanning tree only if it does not form a cycle.
4. Repeat until $n-1$ edges are included.

Key Characteristics

- Partial results may be a forest (disconnected components).
- Requires efficient cycle detection (typically via Disjoint Set / Union-Find).

IV. Prim's Algorithm: Vertex-Based Greedy Approach

Methodology

1. Start with an arbitrary node.
2. Identify edges connecting the current connected component to unexplored nodes.
3. Select the edge with minimum weight.
4. Add the new node and edge.
5. Repeat until all nodes are included.

Key Characteristics

- Partial results are always connected.
- At each step, selects the minimum-cost bridge between explored and unexplored vertices.

V. Advanced Constraints and Comparative Insights

Negative Edge Weights

- Greedy MCST algorithms are not affected by negative edge weights because cycles are not allowed, avoiding infinite loops.

Uniqueness and Cost

Condition	Outcome
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All edge weights are distinct	Unique MCST
Multiple edges have same weight	Multiple MCSTs possible
Cost consistency	Total minimum cost is always identical and unique

